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Heat exchanger with build powder in barrier channels

Abstract

An additively manufactured heat exchanger configured to transfer heat between a first fluid and a second fluid includes a first channel with a first wall configured to port flow of a first fluid and a second channel with a second wall configured to port flow of a second fluid. The heat exchanger also includes a barrier channel containing unprocessed build powder provided by the additive manufacturing process and is located between the first wall and the second wall. The barrier channel is configured to prevent mixing of the first fluid and the second fluid when one of the first wall and the second wall ruptures.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4972902	12/1989	Ninomiya	N/A	N/A
5600057	12/1996	Hansche et al.	N/A	N/A
5772113	12/1997	Gerstmann et al.	N/A	N/A
9746257	12/2016	Fennessy	N/A	N/A
10168114	12/2018	Zager et al.	N/A	N/A
10300530	12/2018	Hart et al.	N/A	N/A
10323887	12/2018	Fennessy	N/A	N/A
10830174	12/2019	Mook et al.	N/A	N/A
11255614	12/2021	Streeter et al.	N/A	N/A
2008/0072425	12/2007	Whittenberger et al.	N/A	N/A
2015/0114611	12/2014	Morris et al.	N/A	N/A
2017/0045313	12/2016	Fennessy	N/A	N/A
2017/0232670	12/2016	Joerger et al.	N/A	N/A
2017/0336155	12/2016	Fennessy	N/A	N/A
2018/0347468	12/2017	Caimano et al.	N/A	N/A
2019/0249926	12/2018	Harris	N/A	N/A
2021/0033354	12/2020	Streeter et al.	N/A	N/A
2021/0071959	12/2020	Streeter et al.	N/A	N/A
2021/0254896	12/2020	Borghese et al.	N/A	N/A
2021/0302109	12/2020	Colson et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1117148	12/1960	DE	N/A
3128497	12/1982	DE	N/A
1998131	12/2007	EP	N/A
2963267	12/2015	EP	N/A
3193123	12/2016	EP	N/A
3284996	12/2017	EP	N/A

3421916	12/2018	EP	N/A
2519411	12/2014	GB	N/A
WO-2017052798	12/2016	WO	F28D 9/0037

OTHER PUBLICATIONS

Extended European Search Report for EP Application No. 19215632.1, Dated Jul. 28, 2020, pp. 9. cited by applicant

First Communication Pursuant to Article 94(3) EPC for EP Application No. 19215632.1, Dated Oct. 28, 2022, pp. 5. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a divisional of U.S. application Ser. No. 16/563,414 filed Sep. 6, 2019 for “HEAT EXCHANGER WITH BUILD POWDER IN BARRIER CHANNELS,” which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) The present disclosure relates generally to thermal management. More specifically, this disclosure relates to heat exchangers in an aircraft having a gas turbine engine.

(2) Heat exchangers are known in the aviation arts and in other industries for providing a compact, low-weight, and highly-effective means of exchanging heat from a hot fluid to a cold fluid. Heat exchangers that operate at elevated temperatures often have short service lives and/or require increased maintenance as a result of high cyclic thermal stress. The stress is caused by multiple system and component factors including rapid flow and/or temperature transients, physical shock, and so on. Stress loading can cause failure of the fluid boundary separating the hot and the cold fluids, thereby resulting in inter-stream contamination of one working fluid by the other. Inter-stream contamination can be disadvantageous, particularly if one of the working fluids is toxic or flammable.

(3) Heat exchanger designs that attempt to reduce the incidence of inter-stream contamination are known in the art, with examples being double closure bars and “inactive” fin or gap layers in plate-fin heat exchangers, and double tube sheets in shell-tube heat exchangers. However, there is a need for a robust heat exchanger design that can reduce inter-stream contamination in the event of the failure of a heat exchanger core component.

SUMMARY

(4) An additively manufactured heat exchanger configured to transfer heat between a first fluid and a second fluid includes a first channel with a first wall configured to port flow of a first fluid and a second channel with a second wall configured to port flow of a second fluid. The heat exchanger also includes a barrier channel containing unprocessed build powder provided by the additive manufacturing process and is located between the first wall and the second wall. The barrier channel is configured to prevent mixing of the first fluid and the second fluid when one of the first wall and the second wall ruptures.

(5) A method for additively manufacturing a heat exchanger having a first channel, a second channel, and a barrier channel includes building a first wall defining the first channel having and configured to port flow of a first fluid, in a layer by layer process and building a second wall defining the second channel having and configured to port flow of a second fluid, in a layer by

layer process. The method includes defining the barrier channel by the first wall and the second wall to prevent mixing of the first fluid and the second in the event of a rupture of either the first wall or the second wall and leaving unprocessed build powder in the barrier channel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a top view of a heat exchanger with barrier passages.
- (2) FIG. 1B is a perspective view of the heat exchanger shown in FIG. 1A.
- (3) FIG. 1C is a front view of the heat exchanger shown in FIG. 1A.
- (4) FIG. 1D is a side view of the heat exchanger shown in FIG. 1A.
- (5) FIG. 2A is the top view of the heat exchanger shown in FIG. 1A showing cross-sectional cut line 2B-2B.
- (6) FIG. 2B is a cross-sectional front view showing the core area of the heat exchanger shown in FIG. 2A taken at cut line 2B-2B.
- (7) FIG. 2C is an enlarged cross-sectional front view showing four hot flow assemblies shown in FIG. 2A.
- (8) FIG. 3A is a cross-sectional front view showing a second embodiment of four hot flow assemblies.
- (9) FIG. 3B is a cross-sectional front view showing a third embodiment of four hot flow assemblies.
- (10) FIG. 3C is a cross-sectional front view showing a fourth embodiment of four hot flow assemblies.
- (11) FIG. 4A is a cross-sectional front view showing a fourth embodiment with seven hot flow assemblies.
- (12) FIG. 4B is the side view of the heat exchanger in FIG. 4A showing cross-sectional cut line 8A-8A.

DETAILED DESCRIPTION

- (13) As aircraft engine efficiency has been improved over the years, there has been a general trend toward generating hotter temperatures in the engine core. Even when materials are used with higher heat tolerances, the parts within the engine core may still experience thermal fatigue over time. There is a need to effectively manage these temperatures in order to maximize the useful life of engine core parts.
- (14) A heat exchange device is disclosed herein, which effectively transfers thermal heat from a hot source to a heat sink within an aircraft engine while also meeting the stringent safety requirements of regulatory agencies. For example, ammonia may be a preferred refrigerant, but can be toxic if introduced into cabin air within the aircraft environmental control system. As such, it is critical to keep ammonia from mixing with air in an environmental control system during operation of the heat exchanger.
- (15) A barrier chamber is provided within the disclosed heat exchanger, which prevents mixing of the two fluids even in the event of a rupture of one of the flow channels of either the first fluid or the second fluid. This redundancy of the barrier chamber in addition to the walls of each flow channel provides an increased operating safety margin for the heat exchanger.
- (16) FIG. 1A is a top view of a heat exchanger with barrier passages. FIG. 1B is a perspective view of the heat exchanger shown in FIG. 1A. FIG. 1C is a front view of the heat exchanger shown in FIG. 1A. FIG. 1D is a side view of the heat exchanger shown in FIG. 1A. Shown in FIGS. 1A-1D are heat exchanger **10**, hot inlet **12**, hot outlet **14**, cold inlet **16**, cold outlet **18**, inlet plenum **20**, inlet transition region **22**, core **24**, outlet transition region **26**, outlet plenum **28**, hot layer **30**, cold layer **32**, and hot inlet port **34**. Core length **L** is labeled in FIG. 1A, and the numbers of vertical layers

N.sub.V and horizontal channels N.sub.H are labeled in FIG. 1B.

(17) Hot inlet flow such as, for example the first fluid formed of bleed air, enters heat exchanger **10** at hot inlet **12**, and the hot outlet flow exits heat exchanger **10** at hot outlet **14**. The cold inlet flow such as, for example fuel, oil, or ammonia enters heat exchanger **10** at cold inlet **16**, and the cold outlet flow exits heat exchanger **10** at cold outlet **18**. In the illustrated embodiment, the hot fluid enters hot inlet **12** and exits hot outlet **14** at a direction that is about perpendicular to the direction of the cold fluid flowing from cold inlet **16** to cold outlet **18** through heat exchanger **10**. Hot fluid flows through inlet plenum **20** which changes the hot flow direction about 90 degrees. Accordingly, the hot fluid flows counter-parallel to the cold fluid through core **24**. In other embodiments the hot fluid can flow parallel to the cold fluid through core **24**. After flowing through inlet plenum **20**, the hot fluid flows through inlet transition region **22**, core **24**, outlet transition region **26**, and through outlet plenum **28**. In the illustrated embodiment, outlet plenum **28** changes the flow direction of the hot fluid about 90 degrees before exiting at hot outlet **14**.

(18) Heat transfer between the hot and cold fluid occurs throughout heat exchanger **10**, including inlet plenum **20**, inlet transition region **22**, core **24**, outlet transition region **26**, and outlet plenum **28**. Core length L characterizes the length of core **24**, where the majority of heat transfer occurs in heat exchanger **10**. An appreciable amount of heat transfer can also occur across inlet and outlet plenums **20**, **28**, and some heat transfer can also occur across outlet transition regions **22**, **26**. The relative sizes of core **24**, inlet and outlet plenums **20**, **28**, and inlet and outlet transition regions **22**, **26** can affect the amount of heat transfer that occurs in these corresponding areas in heat exchanger **10**.

(19) Heat exchanger **10** includes alternating hot layers **30** and cold layers **32**, with any two vertically-adjacent hot layers **30** defining a cold layer **32** therebetween. The number of vertical layers N.sub.V refers to the number of hot layers **30** in heat exchange **10**. In the illustrated embodiment, there are five hot layers **30**, thereby defining four cold layers **32** therebetween. Each of cold layers **32** provides either a counter-parallel or a parallel flow path for cold fluid through heat exchanger **10**. In some embodiments, hot layers **30** and cold layers **32** configurations are selected from the group consisting of plate fin, tube bundle, cross-flow, counter-flow, and multiple pass cross-counter-flow.

(20) Heat exchanger **10** shown in FIGS. 1A-1D can be referred to as a “hot exterior” heat exchanger, because cold fluid does not flow exterior to the outermost hot layers **30**. In some embodiments, cold fluid can flow exterior to the outermost hot layers **30**. In any particular embodiment, the number of hot layers **30** will generally be similar to the number of cold layers **32**. In other embodiments, heat exchanger **10** can have any number of hot and cold layers **30**, **32**.

(21) As shown in FIG. 1B, the number of horizontal channels N.sub.H refers to the number of hot inlet ports **34** in each hot layer **30** in inlet plenum **20**. In the illustrated embodiment, there are ten hot inlet ports **34** in each hot layer. Hot inlet **12** and hot outlet **14** generally have a rectangular envelope and provide the same number of hot inlet ports **34** in each hot layer **30**. Accordingly, heat exchanger **10** can be described as having a rectangular cuboid geometrical shape. In some embodiments, hot inlet **12** and/or hot outlet **14** can have a non-rectangular geometrical configuration, with non-limiting examples being triangular and trapezoidal. In these other embodiments, the number of horizontal channels N.sub.H can be different across various hot layers **30**.

(22) The hot fluid can be referred to as a first fluid, and the cold fluid can be referred to as a second fluid. Accordingly, hot inlet **12** can be referred to as the first fluid inlet, hot outlet **14** can be referred to as the first fluid outlet, cold inlet **16** can be referred to as the second fluid inlet, cold outlet **18** can be referred to as the second fluid outlet, hot layers **30** can be referred to as the first fluid layers, cold layers **32** can be referred to as the second fluid layers, and hot inlet port **34** can be referred to as the first fluid inlet port. Hot and/or cold layers **30**, **32** can also be referred to as hot and/or cold channels, respectively.

(23) As shown in FIG. 1A, heat exchanger core **24** as viewed from the top has an outer geometric shape similar to a parallelogram. Accordingly, core length L can be said to be equivalent to the base of the parallelogram. In the illustrated embodiment, core length L is about 6 inches (15 cm). In some embodiments, core length L can range from about 2 inches (5 cm) to 24 inches (61 cm). In other embodiments, core length L can be less than 2 inches (5 cm) or more than 24 inches (61 cm). It is to be appreciated that in some embodiments of heat exchanger core **24**, core length L can scale with the height and width of heat exchanger core **24**. In a particular embodiment of heat exchanger core **24**, core length L can be zero or nearly zero. Therefore, in that particular embodiment, all or nearly all of the heat transfer will take place across inlet and outlet plenums **20**, **28**, and some heat transfer can also occur across outlet transition regions **22**, **26**.

(24) Heat exchanger **10** including core **24**, inlet and outlet plenums **20**, **28**, and inlet and outlet transition regions **22**, **26** can be formed by an additive manufacturing process. In an exemplary embodiment, powder bed fusion using techniques such as laser sintering, electron beam melting, or glue binder jetting can be used in an additive manufacturing process to fabricate heat exchanger **10** from metallic materials. In some embodiments, unprocessed powder (for example, un-sintered or un-melted powder) is left in the barrier channels shown in FIGS. 2B and 2C. Leaving unprocessed powder in the barrier channels eases manufacturing by eliminating the step of removing the unprocessed powder. Additionally, the barrier channels can be made much smaller compared to a heat exchanger where the unprocessed powder is removed from the barrier channels.

(25) Non-limiting examples of metallic materials that can be used include nickel, aluminum, titanium, copper, iron, cobalt, and alloys thereof. In some embodiments, various alloys of Inconel™ can be used to fabricate heat exchanger **10**, such as Inconel™ **625** and Inconel™ **718**. In other embodiments, Haynes™ **282** can be used in fabricating heat exchanger **10**. In yet other embodiments, alloys of aluminum such as, for example A**205**, can be used in fabricating heat exchanger **10**.

(26) FIG. 2A is the top view of heat exchanger **10** shown and described above in FIG. 1A, showing cross-sectional cut line 2B-2B. FIG. 2B is a cross-sectional front view showing the core area of heat exchanger **10** shown in FIG. 2A taken at cut line 2B-2B. FIG. 2C is an enlarged cross-sectional front view showing four hot flow assemblies shown in FIG. 2B. Shown in FIGS. 2B-2C are hot layers **30**, cold layers **32**, hot flow assemblies **40**, inner channels **42**, inner channel walls **44**, barrier channels **46**, barrier channel walls **48**, vertical vanes **50**, horizontal vanes **52**, and barrier channel vanes **54**.

(27) FIG. 2C also shows horizontal spacing H , vertical spacing V , cold channel height C , horizontal vane thickness $T_{\text{sub.H}}$, vertical vane thickness $T_{\text{sub.V}}$, inner channel diameter $D_{\text{sub.I}}$, inner channel wall thickness $T_{\text{sub.I}}$, barrier channel diameter $D_{\text{sub.B}}$, and barrier channel wall thickness $T_{\text{sub.B}}$. The cross-sectional view of core **24** shown in FIG. 2B depicts five hot layers **30** (where the number of vertical layers $N_{\text{sub.V}}=5$), each having ten hot flow assemblies **40** (where the number of horizontal layers $N_{\text{sub.H}}=10$). A cold layer **32** is sandwiched between any two adjacent hot layers **30**. In the illustrated embodiment, each hot flow assembly **40** corresponds to an associated hot inlet port **34**. In some embodiments, two or more hot flow assemblies **40** can correspond to an associated hot inlet port **34**. In other embodiments, one hot flow assembly **40** can correspond to two or more associated hot inlet ports **34**.

(28) As shown in FIGS. 2B-2C, each hot layer **30** within core **24** includes a number of hot flow assemblies **40**. Vertically-adjacent hot flow assemblies **40** (separated by a single cold layer **32**) from one hot layer **30** to the next can be supported vertically by vertical vanes **50**, and horizontally-adjacent hot flow assemblies **40** within any particular hot layer **30** can be supported horizontally by horizontal vanes **52**. Each hot flow assembly **40** includes inner channel **42** located coaxially interior to an associated barrier channel **46**, supported by barrier channel vanes **54**. Each inner channel **42** is defined by an associated inner channel wall **44**, and each barrier channel **46** is defined by an associated inner channel wall **44** and barrier channel wall **48**.

(29) The hot fluid flows in parallel through the various inner channels **42**, the flow through each inner channel **42** being directed by an associated hot inlet port **34**. Accordingly, inner channel walls **44** provide a pressure boundary for the hot fluid. The cold fluid flows through respective cold layers **32**, with barrier channel walls **48** providing a pressure boundary for the cold fluid.

(30) In some embodiments, horizontal spacing H can range from about 0.1 inch (2.5 mm) to 1 inch (25 mm). In other embodiments, horizontal spacing H can be less than 0.1 inch (2.5 mm) or greater than 1 inch (25 mm). In some embodiments, vertical spacing V can range from about 0.1 inch (2.5 mm) to 1 inch (25 mm). In other embodiments, vertical spacing V can be less than 0.1 inch (2.5 mm) or greater than 1 inch (25 mm). In some embodiments, cold channel height C can range from about 0.05 inch (1.3 mm) to 0.5 inch (13 mm). In other embodiments, cold channel height C can be less than 0.05 inch (1.3 mm) or greater than 0.5 inch (13 mm). In some embodiments, inner channel diameter $D_{sub.I}$ can range from about 0.05 inch (1.3 mm) to 0.5 inch (13 mm). In other embodiments, inner channel diameter $D_{sub.I}$ can be less than 0.05 inch (1.3 mm) or greater than 0.5 inch (13 mm). In some embodiments, barrier channel diameter $D_{sub.B}$ can range from about 0.08 inch (2.0 mm) to 0.5 inch (13 mm). In other embodiments, barrier channel diameter $D_{sub.B}$ can be less than 0.08 inch (2.0 mm) or greater than 0.5 inch (13 mm).

(31) In some embodiments, horizontal vane thickness $T_{sub.H}$, vertical vane thickness $T_{sub.V}$, and barrier channel wall thickness $T_{sub.B}$ are about the same thickness. Barrier channel vane thickness (not labeled) can be similar in dimension to horizontal and/or vertical vane thicknesses $T_{sub.H}$, $T_{sub.V}$. In some embodiments, horizontal vane thickness $T_{sub.H}$, vertical vane thickness $T_{sub.V}$, and/or barrier channel wall thickness $T_{sub.B}$ can range from about 0.008 inch (0.2 mm) to 0.02 inch (0.5 mm). In other embodiments, horizontal vane thickness $T_{sub.H}$, vertical vane thickness $T_{sub.V}$, and/or barrier channel wall thickness $T_{sub.B}$ can be less than 0.008 inch (0.2 mm) or greater than 0.02 inch (0.5 mm). In any embodiment, some or all of horizontal vane thickness $T_{sub.H}$, vertical vane thickness $T_{sub.V}$, barrier channel wall thickness $T_{sub.B}$, and barrier vane thickness can be different from each other.

(32) In some embodiments, unprocessed build powder is removed from cold layers **32** and inner channels **42**, but left in place in barrier channels **46**. While unprocessed build powder can be removed through the use of rotation, impact, and vibration techniques or by blowing out the powder with a compressed gas source, the barrier channel size and geometry can be limited to allow removal of the build powder using these techniques. For example, a barrier channel without any sharp turns or constricted flow paths may be limited to having a span (the distance from the inner channel wall to the barrier channel wall) of about 0.010 inches (0.25 mm) or more such that the powder may be removed using these techniques. However, a barrier channel with a tortured flow path, for example having sharp turns or constrictions in the flow path, may be limited to having a thickness of about 0.030 inches (0.75 mm) or more. However, if the build powder in the barrier channels is left in place, the thickness of barrier channel **46** can be about 0.008 inches (0.2 mm). In other words, the span between inner channel wall **44** and barrier channel wall **48** can be about 0.008 inches (0.2 mm). In other embodiments the thickness of barrier channel **46** is from 0.005 inches (0.13 mm) to 0.030 inches (0.75 mm), inclusive. In other embodiments, the thickness of barrier channel **46** can be greater than 0.030 inches (0.75 mm).

(33) Leaving the build powder in place in the barrier channels eases the additive manufacturing process by eliminating the process step of removing the build powder from the barrier channels. Additionally, leaving the build powder in place allows a narrower barrier channel to be constructed. Build powder is also a better heat conductor compared to air. As such, a narrower barrier channel containing build powder may allow construction of a heat exchanger capable of exchanging more heat with the same footprint or allow construction of a heat exchanger capable of exchanging the same amount of heat with a smaller footprint compared to a heat exchanger with larger barrier channels filled with air or other matter less conductive than the build powder.

(34) During normal operation of heat exchanger **10**, each barrier channel **46** is devoid of working

fluid. Only in the event of a rupture in either inner channel wall **44** or barrier channel wall **48** will a corresponding working fluid enter barrier channel **46**. In the illustrated embodiment, barrier channels **46** are vented to the ambient. In the event of a material failure within heat exchanger **10**, a leaking working fluid will flow into one or more barrier channels **46**, thereby preventing the inter-stream contamination between the hot and cold fluids. In other words, barrier channels **46** can be described as providing leak prevention or barrier protection with reference to helping prevent mixing between the hot and cold fluid circuits. In some embodiments, drain **456** (shown in FIG. **4B**) can be provided to direct the leaking fluid to a collection point that is observable by an operator. In some embodiments, detector **458** (shown in FIG. **4B**) such as, for example, a pressure sensor can be provided to monitor for leakage into barrier channels **46**.

(35) Referring to FIG. **2C**, heat transfer occurs across each of the hot flow assemblies **40** from the hot fluid within inner channels **42**, through inner channel walls **44**, across barrier channels **46**, and through barrier channel walls **48** into the cold fluid within cold layers **32**. In some embodiments, a portion of the heat transfer occurring within each one hot flow assembly **40** will be by thermal conduction radially through barrier channel vanes **54** from inner channel wall **44** to barrier channel wall **48**. In these embodiments, the relative proportion of heat transfer through barrier channel vanes **54** depends on several factors such as, for example the thermal conductivity of the particular material used and the relative shape, size, and/or number of barrier channel vanes **54**.

(36) The foregoing description is based on the hot fluid having a higher temperature than the cold fluid, with the hot fluid being directed through the inner channels. In some embodiments, the hot fluid can be directed through the cold layers, and the cold fluid can be directed through the inner channels.

(37) A leakage passage can be provided through each plenum vane in the barrier channels, thereby fluidly connecting all barrier channels together. The leakage passage, for example, can be an aperture, hole, or orifice, which allows a fluid to pass between adjacent barrier channels. The leakage passages can lead to a drain, providing a discharge path from heat exchanger **10** for any fluid that accumulates in the barrier channels. The barrier channels are normally devoid of working fluids, but a breach of either the hot or cold fluid boundary will result in the leakage of the respective fluid into one or more barrier channels, thereby preventing inter-stream contamination between the hot and cold fluid circuits. In some embodiments, the drain can be located at or near a gravity low-point of the heat exchanger, thereby providing for the gravity removal of fluid in one or more barrier channels from heat exchanger **10**.

(38) The drain can be configured to provide an indication of the presence of a leak, thereby providing an indication of a rupture of a hot and/or cold pressure boundary within the heat exchanger. The presence of fluid or build powder near the drain may be observable to a user and indicate the presence of a leak in the heat exchanger. A detector can also be placed near the drain to indicate the presence of a leak in the system. For example, a pressure sensor placed in the flow path of the drain can notify a user that a pressure change has occurred, which may be indicative of a leak in the heat exchanger.

(39) FIG. **3A** is a cross-sectional front view showing a second embodiment of four hot flow assemblies, a view that is similar to that shown above in regard to FIG. **2C**. Shown in FIG. **3A** are cold layer **132**, hot flow assembly **140**, inner channel **142**, inner channel wall **144**, barrier channel **146**, barrier channel wall **148**, vertical vanes **150**, horizontal vanes **152**, and barrier channel vanes **154**, all of which have descriptions that are substantially similar to those provided above in regard to FIG. **2C**, with the exception that each particular barrier channel **146** includes eight barrier channel vanes **154**. In the illustrated embodiment, the eight barrier channel vanes **154** are spaced approximately equally around inner channel **142** and in each barrier channel **146**, which connects inner channel wall **144** to barrier channel wall **148**.

(40) In some embodiments, more than or fewer than eight barrier channel vanes **154** can be used in a particular barrier channel **146**, each having radial orientations that are different from those shown

in the illustrated embodiment. Several factors can be considered in the selection of the quantity and alignment of barrier channel vanes **154** such as, for example, the need for greater structural support within the heat exchanger, the containment of relatively high hot fluid and/or cold fluid system pressures, the providing greater thermal conductivity from inner channel wall **144** to barrier channel wall **148** via barrier channel vanes **154**, and additive manufacturing process optimization. In other embodiments, different hot flow assemblies **140** within the same heat exchanger can have different numbers of barrier channel vanes **154** throughout respective barrier channels **146**.

(41) FIG. 3B is a cross-sectional front view showing a third embodiment of four hot flow assemblies, a view that is similar to that shown above in regard to FIG. 2C. Shown in FIG. 3B are cold layer **232**, hot flow assembly **240**, inner channel **242**, inner channel wall **244**, barrier channel **246**, barrier channel wall **248**, vertical vanes **250**, horizontal vanes **252**, and barrier channel vanes **254**. The features shown in FIG. 3B have descriptions that are substantially similar to those provided above in regard to FIG. 2C with the exception that inner channel wall **244** and barrier channel wall **248** have rhomboid cross-sectional geometries. The configuration shown in the illustrated embodiment can be advantageous in certain applications compared to a circular cross-sectional geometry such as, for example, providing greater structural strength and/or stability for the heat exchanger, controlling the stiffness of the heat exchanger along a particular axis, and optimizing the additive manufacturing process.

(42) The configuration shown for inner channel wall **244** and barrier channel wall **248** in the illustrated embodiment uses square rhomboids. In some embodiments, inner channel wall **244** and/or barrier channel wall **248** can use non-square rhomboids such as, for example, kite-shape, parallelogram-shape, and irregular shape. In other embodiments, inner channel wall **244** and barrier channel wall **248** geometries can be different from each other. For example, in a particular embodiment, inner channel wall **244** can have a circular geometry and barrier channel wall **248** can have a rhomboid geometry.

(43) FIG. 3C is a cross-sectional front view showing a fourth embodiment of four hot flow assemblies, a view that is similar to that shown above in regard to FIG. 2C. Shown in FIG. 3C are cold layer **332**, hot flow assembly **340**, inner channel **342**, inner channel wall **344**, barrier channel **346**, barrier channel wall **348**, vertical vanes **350**, horizontal vanes **352**, and barrier channel vanes **354**. The features shown in FIG. 3C have descriptions that are substantially similar to those provided above in regard to FIG. 2C with the exception that inner channel wall **244** and barrier channel wall **248** have elliptical geometries. Also labeled in FIG. 3C are inner channel major axis length $E_{sub.I}$, inner channel minor axis length $F_{sub.I}$, barrier channel major axis length $E_{sub.B}$, and barrier channel minor axis length $F_{sub.B}$. In the illustrated embodiment, the major axis of inner channel **342** is defined by inner channel major axis length $E_{sub.I}$ and barrier channel **346** is defined by barrier channel major axis length $E_{sub.B}$.

(44) An inner channel aspect ratio can be defined as the ratio of inner channel major axis length $E_{sub.I}$ to inner channel minor axis length $F_{sub.I}$. In the illustrated embodiment, the inner channel aspect ratio is about 1.25. In some embodiments, the inner channel aspect ratio can range from 1 to about 2.5. In other embodiments, the inner channel aspect ratio can be greater than 2.5. In an embodiment where inner channel major axis length $E_{sub.I}$ is oriented horizontally, then inner channel **342** can be said to have a particular aspect ratio that is oriented horizontally.

(45) A barrier channel aspect ratio can be similarly defined as the ratio of barrier channel major axis length $E_{sub.B}$ to barrier channel minor axis length $F_{sub.B}$. In the illustrated embodiment, the barrier channel aspect ratio is about equal to the inner channel aspect ratio. As the inner channel aspect ratio, the barrier channel aspect ratio can range from 1 to about 2.5. In other embodiments, the inner channel aspect ratio can be greater than 2.5. In some embodiments, the inner channel aspect ratio and the barrier channel aspect ratio can be different values from each other. For example, in a particular embodiment, inner channel **342** can be circular, having an inner channel aspect ratio of 1, and the barrier channel aspect ratio can be greater than 1.

(46) In other embodiments, major axes of inner channel **342** and/or barrier channel **346** can be oriented in a direction that is other than horizontal or vertical. In other embodiments, the major axes orientations and the aspect ratios of inner channels **242** and/or barrier channels **346** can vary from one hot flow assembly **340** to another throughout a particular heat exchanger. In other embodiments, the cross-sectional shapes of inner channel **342** and/or barrier channel **346** can be triangular, pentagonal, hexagonal, octagonal, or oval.

(47) FIG. **4A** is a cross-sectional front view showing a fourth embodiment with seven hot flow assemblies, a view that is similar to that shown above in regard to FIG. **2C**. Shown in FIG. **4A** are cold layer **432**, hot flow assembly **440**, inner channel **442**, inner channel wall **444**, barrier channel **446**, barrier channel wall **448**, vertical vanes **450**, horizontal vanes **452**, and barrier channel vanes **454**. The features shown in FIG. **4A** have descriptions that are substantially similar to those provided above in regard to FIG. **2C** with the exception that the flow assemblies **440** are arranged in a hexagonal geometry.

(48) FIG. **4B** is the side view of the heat exchanger in FIG. **4A** showing cross-sectional cut line **4A-4A**. Shown in FIG. **4B** are cold layer **432**, hot flow assembly **440**, inner channel **442**, inner channel wall **444**, barrier channel **446**, and barrier channel wall **448**. In the illustrated embodiment, barrier channel **446** surrounding inner channel **442** are circumferentially related and in fluid communication. Additionally, barrier channels **446** are in fluid communication with one another in adjacent flow assemblies **440**. For example, in the illustrated embodiment, a barrier channel running perpendicular to inner channel **442** can provide fluid communication between adjacent flow assemblies and/or the perpendicular barrier channel can be in fluid communication with drain **456**.

(49) In some embodiments, the additional barrier channels **446** run at an acute angle to the barrier channels surrounding inner channels **442**. Having all barrier channels **446** in fluid communication with a common drain allows a user to identify a potential leak through visual inspection by detecting liquid or unprocessed build powder on the exterior of the heat exchanger or by detector **458** such as a pressure sensor signaling an increase in pressure at drain **456** due to a leak in either inner channel wall **444** or barrier channel wall **448**.

(50) Heat exchanger **10** in the foregoing embodiments is built using additive manufacturing techniques such as, for example, laser sintering, electron beam melting, and glue binder jetting. During the build process, unprocessed build powder is left in the barrier channels and the unprocessed powder is only removed from the inner channels and cold layers.

(51) Leaving the build powder in place in the barrier channels eases the additive manufacturing process by eliminating the process step of removing the build powder from the barrier channels. This allows a narrower barrier channel to be constructed while still providing protection against the mixing of the two fluids when either the inner channel wall or the barrier wall ruptures.

(52) Unprocessed build powder, which can have a density of about 80% compared to processed build powder, is also a better heat conductor compared to air. As such, a narrower barrier channel containing build powder may allow construction of a heat exchanger capable of exchanging more heat with the same footprint or allow construction of a heat exchanger capable of exchanging the same amount of heat with a smaller footprint compared to a heat exchanger with larger barrier channels filled with air or other matter less conductive than the build powder.

(53) Unprocessed build powder present in the barrier channel can also dampen vibrational stress induced in the heat exchanger. In other words, the unprocessed build powder can act as a vibration suppressor under load compared to a barrier channel filled with another material such as air that is less resistant to vibrational forces.

Discussion of Possible Embodiments

(54) The following are non-exclusive descriptions of possible embodiments of the present invention.

(55) An additively manufactured heat exchanger configured to transfer heat between a first fluid

and a second fluid includes a first channel with a first wall configured to port flow of a first fluid and a second channel with a second wall configured to port flow of a second fluid. The heat exchanger also includes a barrier channel containing unprocessed build powder provided by the additive manufacturing process and is located between the first wall and the second wall. The barrier channel is configured to prevent mixing of the first fluid and the second fluid when one of the first wall and the second wall ruptures.

(56) The heat exchanger of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(57) The heat exchanger includes a support member, extending between the first wall and the second wall.

(58) The non-processed build powder is a nickel based super alloy.

(59) The shape of the first channel is selected from the group consisting of trapezoid, rectangle, and annulus.

(60) The span between the first wall and the second wall is from 0.005 inches (0.13 mm) to 0.030 inches (0.75 mm), inclusive.

(61) The unprocessed build powder is configured to allow the first fluid or the second fluid to flow through the barrier channel.

(62) The unprocessed build powder in the barrier channel is configured to dampen vibrational stresses in the heat exchanger compared to the barrier channel filled with another material such as air, which is less resistant to vibrational forces.

(63) The heat exchanger includes a drain in fluid communication with the barrier channel and configured to pool liquid or unprocessed build powder on the exterior of the heat exchanger in the event of a rupture of either the first wall or the second wall.

(64) The pool of liquid or unprocessed build powder is visible by a user.

(65) The heat exchanger includes a drain in fluid communication with the barrier channel and a pressure sensor in fluid communication with the drain.

(66) The pressure sensor is configured to send a signal to a user of an increase in pressure in the event of a rupture of either the first wall or the second wall.

(67) The first channel and the second channel configurations are selected from the group consisting of plate fin, tube bundle, cross-flow, counter-flow, and multiple pass cross-counter-flow.

(68) A method for additively manufacturing a heat exchanger having a first channel, a second channel, and a barrier channel includes building a first wall defining the first channel having and configured to port flow of a first fluid, in a layer by layer process and building a second wall defining the second channel having and configured to port flow of a second fluid, in a layer by layer process. The method includes defining the barrier channel by the first wall and the second wall to prevent mixing of the first fluid and the second in the event of a rupture of either the first wall or the second wall and leaving unprocessed build powder in the barrier channel.

(69) The method of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(70) The method includes building a support member, extending between the first wall and the second wall.

(71) The non-processed build powder is a nickel based super alloy.

(72) The shape of the first channel is selected from the group consisting of trapezoid, rectangle, and annulus.

(73) The span between the first wall and the second wall is from 0.005 inches (0.13 mm) to 0.030 inches (0.75 mm), inclusive.

(74) The unprocessed build powder is configured to allow the first fluid or the second fluid to flow through the barrier channel.

- (75) The unprocessed build powder in the barrier channel is configured to dampen vibrational stresses in the heat exchanger compared to the barrier channel filled with another material such as air, which is less resistant to vibrational forces.
- (76) The heat exchanger includes manufacturing a drain in fluid communication with the barrier channel and configured to pool liquid or unprocessed build powder on the exterior of the heat exchanger in the event of a rupture of either the first wall or the second wall.
- (77) The pool of liquid or unprocessed build powder is visible by a user.
- (78) The heat exchanger includes manufacturing a drain in fluid communication with the barrier channel and a pressure sensor in fluid communication with the drain.
- (79) The pressure sensor is configured to send a signal to a user of an increase in pressure in the event of a rupture of either the first wall or the second wall.
- (80) The first channel and the second channel configurations are selected from the group consisting of plate fin, tube bundle, cross-flow, counter-flow, and multiple pass cross-counter-flow.
- (81) While the invention has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment(s) disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. A method for additively manufacturing a heat exchanger configured to transfer heat between a first fluid and a second fluid, the method comprising: providing an alternating plurality of hot layers and a plurality of cold layers, wherein each of the plurality of hot layers provides a flow path through the heat exchanger for the first fluid and is separated from each other of the plurality of hot layers by one of the plurality of cold layers and each of the plurality of cold layers provides a flow path through the heat exchanger for the second fluid, wherein: each of the plurality of hot layers comprises a plurality of hot flow assemblies, and each of the plurality of hot flow assemblies is built by a layer by layer process using additive manufacturing techniques that comprises the steps of: building an inner channel wall that defines an inner channel configured to port flow of the first fluid; and building a barrier channel wall surrounding the inner channel wall such that the barrier channel wall defines a barrier channel coaxially exterior to the inner channel and located between the inner channel wall and the barrier channel wall, wherein the barrier channel is configured to prevent mixing of the first fluid and the second fluid when one of the inner channel wall and the barrier channel wall ruptures; wherein the method further comprises leaving in the barrier channel unprocessed build powder used by the additive manufacturing techniques.
2. The method of claim 1, further comprising building, with a layer by layer process using additive manufacturing techniques, a barrier channel vane extending between the inner channel wall and the barrier channel wall.
3. The method of claim 1, wherein the unprocessed build powder is a nickel based super alloy.
4. The method of claim 1, wherein the shape of the inner channel is selected from the group consisting of trapezoid, rectangle, and annulus.
5. The method of claim 1, wherein the span between the inner channel wall and the barrier channel wall is from 0.005 inches (0.13 mm) to 0.030 inches (0.75 mm), inclusive.
6. The method of claim 1, wherein the unprocessed build powder is configured to allow the first fluid or the second fluid to flow through the barrier channel.
7. The method of claim 1, wherein the unprocessed build powder in the barrier channel is configured to dampen vibrational stresses in the heat exchanger compared to the barrier channel

filled with another material that is less resistant to vibrational forces.

8. The method of claim 1, further comprising building a drain in fluid communication with the barrier channel and configured to pool liquid or unprocessed build powder on the exterior of the heat exchanger in the event of a rupture of either the inner channel wall or the barrier channel wall.

9. The method of claim 8, wherein the pool of liquid or unprocessed build powder is visible by a user.

10. The method of claim 1, further comprising building a drain in fluid communication with the barrier channel and a pressure sensor in fluid communication with the drain.

11. The method of claim 10, wherein the pressure sensor is configured to send a signal to a user of an increase in pressure in the event of a rupture of either the inner channel wall or the barrier channel wall.

12. The method of claim 1, wherein the hot layer and the cold layer configurations are selected from the group consisting of plate fin, tube bundle, cross-flow, counter-flow, and multiple pass cross-counter-flow.
